A 6PROGSDATS Final Document Project

Enrollment System for SOC

Presented to the School of Computing

Holy Angel University

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Project Title: Enrollment System for SOC

Project Description:

The project is about an Enrollment System solely dedicated for SOC students starting from 1st year to 4th year (Except for Cyber Security since it goes up to 5 years). The program will interview the student, asks their name, requirements, scholarship status, dorms, programs from 1st year 1st semester to 5th year 3rd semester. This also includes a payment system, and lastly, an e-portfolio where all the valuable information will be stored, this file can be printed out individually and given to the students. The person who intends to use this is an advisor inputting information coming from the student.

Variables and Methods used:

Database Class (Constructor):

Variables (each has getter methods)

- private int studentAge – sets the age of the student.
- private String name – sets the name of the student.
- private int pays – sets the amount the student paid.
- private String pa – sets the condition if they have parent's approval if they are too young.
- private int sem – sets the semester in which they are enrolling in.
- private int year – sets the year in which they are enrolling in.

- Database() – the constructor, it connects itself to the main program.

Enrollment Class (Main program):

Methods and Variables (each method and variable is defined)

public static void main () – Runs all the methods accordingly.

- Contains Arraylist Container 1 (Database method), 2 (Interview answers), and 3 (Chosen program's subjects)

public static void Database (container 1) – Asks for the requirements

-Uses try-catch looping, Exception Handling, Nested conditional-looping

Variables:

- int askAge = 0; - Asks the age, sets it to the constructor
- String parentApproval = "null"; - asks for parent's approval if below the required age.
- database hau = new database(); - connecting the constructor.java file to the main program.
- String bc = "null"; - Asks for birth certificate
- String bc1 = "null"; - Will be used in the looping until valid answers are given.
- String certificate = "null"; - Asks for shs graduation certificate
- String certificate1 = "null"; Will be used in the looping until valid answers are given.
- String name; - Asks for the name then sets it.
- ArrayList requirements – Is used as a partial container which will be passed to container 1.

Note: Every answer will be recorded in the ArrayList, then every setter is partnered with a getter from the constructor which will be depending on the condition.

public static void Interview (container 2) – Mini-interview if the student wants to apply for scholarship and if they want to use a dorm.

-Uses Nested conditional-looping statements

Variables:

- String scholarship = "null"; - Asks the student if they want to take a scholarship.
- String dorm; - Asks the student if they live outside of Angeles.
- String ask = "null"; - Is used as a condition in a while loop to get a valid answer.
- String ask1; - Asks the student if they want to live in a dorm.
- int age = 0; - Age confirmation before applying for a dorm.
- ArrayList Dorms – Is used as a partial container which will be passed to container 1.
  Note: Every answer will be recorded in the ArrayList.

public static void Program (container 3) – Part of the program where it will ask the user what program they are enrolling for, and then contains it all on container 3.

-Uses Nested conditional statement, setters for year and semester.

Variables:

- database soc = new database(); - connects it to the constructor.java file.
- String course; - Asks for the course from the list, either Network Administrator, Web Development, Computer Science, Cyber Security, and Multimedia.
- int sem = 0; - Asks for which sem the student is enrolling in then sets it in the constructor.
- int year = 0; - Asks for which year the student is enrolling in then sets it in the constructor.
- String sem1 = Is used to convert the setSem to a string and become its holder.
- String year1 = Is used to convert the setYear to a string and become its holder.
- Every result, depending on the condition will be added to the third container from the main method.
  Note: Every program consists of 4 years and 2 semesters each except for Cyber Security, which consists of 5 years and becomes a tri-sem starting from 4th year.

public static void Payment () – Asks the user to pay the PHP 200 processing fee.

-Nested conditional-looping statements, setter for amount of payment.

Variables:

- database it = new database(); - connects it to the constructor.java file.
- String answer1; - Asks the user if they want to pay now.
- String answer2; - If the paid price is below the price then they will be asked if they want to follow it up.

- int price = 200; - Final initiated price for the reservation fee.
- int paid = 0; - Asks the user how much the student paid, it will be put to the setter.
- Int total = (price – paid getter) - this variable will compute if the paid is enough for the price using the price minus paid getter. Note: Everything is conditional, the flow of the program depends on the user.

public static void Portfolio (container 1, container 2, container 3) – Uses all the containers from the main method to be used on the file writer here.

-Try-catch block, timers, FileWriter

Variables:

Timer counter = new Timer(); - Used to set the existence of the timer.

TimerTask task = new TimerTask() – Used to set the task timer of what it will do.

int currentNum = 1; - Start of the timer.

int passed = 6; - The timer's end, it will break once it reaches a number before this.

int j = 1; - This is used to print out the elements on the first container.

int k = 1; -This is used to print out the elements on the second container.

int i = 1; - This is used to print out the elements on the third container.

Note: Every printing of the containers are all inside the try-catch block

Scope and Limitations:

The scope of the program starts from the constructor, main program, Database method, Interview method, Program method, Payment method, and lastly, the Portfolio method. The constructor includes the setters and getters, main program holds everything. Database method covers the name, age, and requirements for enrollment, interview covers the scholarship and dorm questions and answers, the program covers all the subjects from 1st year to last year, every semester, and every program in SOC. The Payment method only covers the processing fee, and lastly, the portfolio covers the printing out of the student profile.

The limitation of this program consists of the following: The unexpected user-errors, unexpected errors from the program itself that we weren't able to see, an SQL database which will add all the student information, sudden changes with the subjects in the programs, the payment method isn't directly connected to the student's tuition, the portfolio may need some improvements.

User Manual:

Database method

```
Enter your name: Juan Paolo Vivaz
Input your age: 34
Do you acquire a Birth Certificate? (Yes/No): Yes
Do you acquire Senior High School Certificate of graduation? (Yes/No): Yes
```

Once the program starts, it will start with the first method – the Database method.
In this method, the user will be tasked to write their name, age, and some questions related to the required documents.

Interview method

```
Are you considering to take a scholarship? (Yes/No): No
Do you live outside of Angeles City? (Yes/No): Yes
Would you like to use a dorm? (Yes/No): No
If you reconsider, feel free to visit the office of dormitory
```

After the Database method, the user will then proceed to the Mini-interview. This will ask them if they want to take a scholarship, if they live outside of Angeles City, and if they would like to use a dorm.

Program method

```
A = Network Admintrator
B = Web Development
C = Computer Science
D = Cyber Security
E = MultiMedia
Enter your program: B
Enter what year you are enrolling in (1|2|3|4): 2
Enter which semester you are enrolling in (1|2): 1
The needed courses for your program are:
Minor Subjects:
-PE3 (2 units)
-Ethics (3 units)
-Calculus (3 units)

Major Subjects:
-Object-Oriented Programming  (3 units)
-Operating Systems Application (3 units)
-Data Structures and Algorithms (3 units)
-Introduction to Web Programming (3 units)
-Ethics for Computing Professionals (3 units)
A total of 23 units
```

After the Mini-interview, the student will be given the options to choose between the programs available, they will be tasked to enter which program by indicating the letter, enter what year they will be enrolling, and what semester they will be enrolling. After all that, the subjects for the given information will be printed out.

Payment method

```
The reservation/processing fee of your enrollment is worth PHP200.
Would you like to pay now? (Yes/No): yes
Enter the amount they paid: 190
Your remaining balance is: 10
Would you like to follow-up the remaining balance? (Yes/No): yes
Your remaining balance will be added to your student tuition. Thank you for enrolling! Please wait while the teller prints out your student e-portfolio.
```

After choosing which program they are going to study, they will proceed to the payment system, they will be tasked to pay an amount to cover the processing fee, they can also either pay it now or pay it later.

Portfolio method

```
Processing... 1

Processing... 2
Processing... 3
Processing... 4
Processing... 5
Congratulations! You're officially enrolled! Here is a copy of your student portfolio
```

After all that, now for the processing of the portfolio, the user will have to wait for it to process. Then they will be enrolled to the school. They will be given an E-Portfolio which is a sign of successful enrollment.

Whole program overview

```
C:\Users\srjee\Downloads\finals>javac enrollment.java

C:\Users\srjee\Downloads\finals>javac database.java

C:\Users\srjee\Downloads\finals>java enrollment.java
Enter your name: Juan Paolo Vivaz
Input your age: 34
Do you acquire a Birth Certificate? (Yes/No): Yes
Do you acquire Senior High School Certificate of graduation? (Yes/No): Yes


Are you considering to take a scholarship? (Yes/No): No
Do you live outside of Angeles City? (Yes/No): Yes
Would you like to use a dorm? (Yes/No): No
If you reconsider, feel free to visit the office of dormitory


A = Network Admintrator
B = Web Development
C = Computer Science
D = Cyber Security
E = MultiMedia
Enter your program: B
Enter what year you are enrolling in (1|2|3|4): 2
Enter which semester you are enrolling in (1|2): 1
The needed courses for your program are:
Minor Subjects:
-PE3 (2 units)
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Major Subjects:
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-Data Structures and Algorithms (3 units)
-Introduction to Web Programming (3 units)
-Ethics for Computing Professionals (3 units)
A total of 23 units


The reservation/processing fee of your enrollment is worth PHP200.
Would you like to pay now? (Yes/No): yes
Enter the amount they paid: 190
Your remaining balance is: 10
Would you like to follow-up the remaining balance? (Yes/No): yes
Your remaining balance will be added to your student tuition. Thank you for enrolling! Please wait while the teller prints out your student e-portfolio.


Processing... 1

Processing... 2
Processing... 3
Processing... 4
Processing... 5
Congratulations! You're officially enrolled! Here is a copy of your student portfolio
```
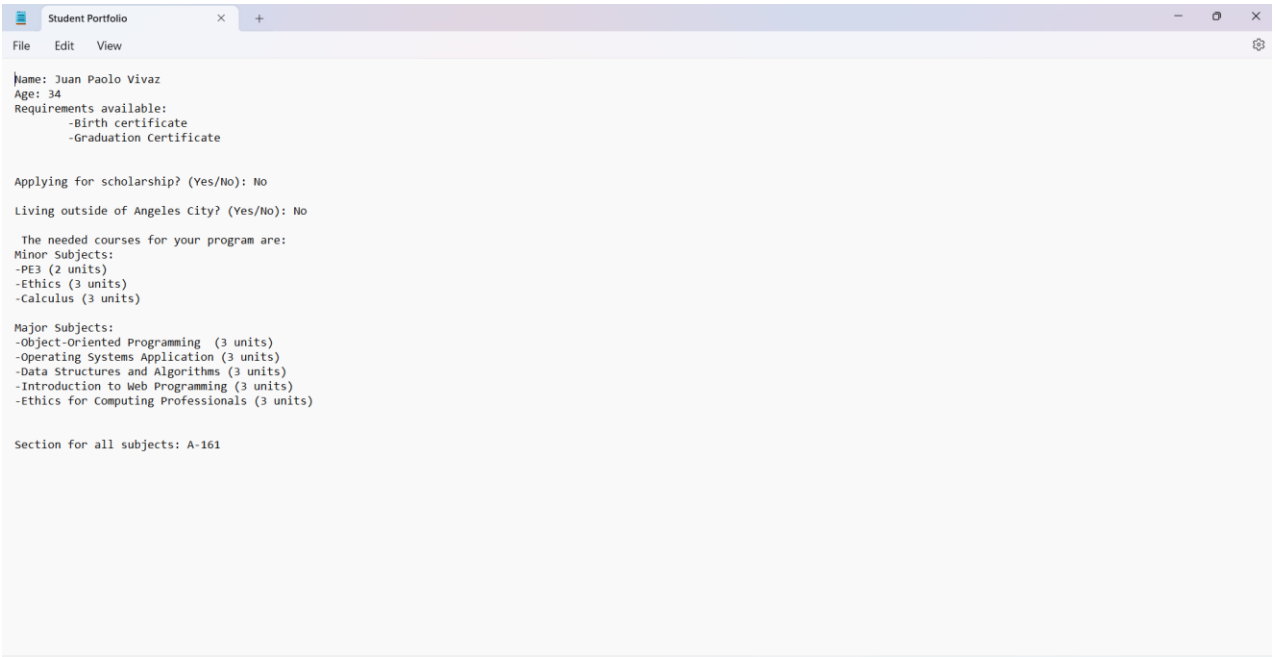

This is the whole program from start to finish, it is full of conditional looping statement so it will differ from what the user will enter.

E-portfolio sample from the test



```
Name: Juan Paolo Vivaz
Age: 34
Requirements available:
        -Birth certificate
        -Graduation Certificate


Applying for scholarship? (Yes/No): No

Living outside of Angeles City? (Yes/No): No

 The needed courses for your program are:
Minor Subjects:
-PE3 (2 units)
-Ethics (3 units)
-Calculus (3 units)

Major Subjects:
-Object-Oriented Programming  (3 units)
-Operating Systems Application (3 units)
-Data Structures and Algorithms (3 units)
-Introduction to Web Programming (3 units)
-Ethics for Computing Professionals (3 units)


Section for all subjects: A-161
```

This is the student e-portfolio which contains all the information the student has based on what the student submitted.

Team Workload Distribution:

Jeerh, Satwinder R.	Lamorena, John Ezekiel
Documentation - Description - Classes, Variables and Methods Used Program - Java file for setters and getters, Database method, and raw working plan of the program General - Reporter, writer	Documentation - Sample outputs Program - Mini-interview method General - Reporter
Venasquez, John Manuel	Venzon, Mathew
Documentation - Scope of the study Program - Program method - main method General - Reporter, video editor	Documentation - Limitation of the study Program - Payment method and clean up of other methods General - Ppt editor, reporter, video filming
Vivz, Juan Paolo	
Documentation - Project Idea - Project title - Documentation editor Program - Portfolio method and clean up of the whole program General - Supervised the whole process, reporter, advisor	

References:

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